

Stablecoin Yield

Persistence, mechanisms and observed TVL response

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Revised analytical report

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The empirical analysis retains the historical sample through 8 July 2026: 250 pools and 137,095 pool-days. This edition adds explicit tool attribution, an offline interactive companion, reproducibility instructions and statistical interpretation notes.

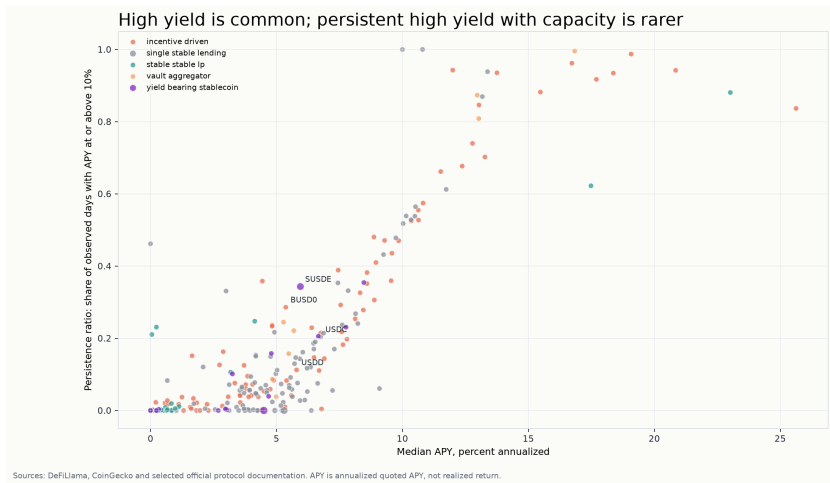
Reading guide. The following pages preserve the original analytical report. Appendix A documents the revision and qualifies the uncertainty and reproducibility claims. Read the two parts together.

Code repository

<https://github.com/Laimon99/stablecoin-yield-visualization>

Original code: MIT. Original presentation, report and text: [CC BY 4.0](#), Simone Ragusini (2026). Provider data and third-party dependencies retain their own terms. See LICENSE and NOTICE.md.

Educational descriptive analysis. Quoted APY is not realized return. The project does not identify a safest or best investment.



Stablecoin yield joint screen

250
pools

137,095
pool-days

2.0
median
high-yield days

The price of yield

Persistence, mechanisms and TVL
response in stablecoin DeFi

A visual analytics report on why quoted
stablecoin APY should be read as a regime with
duration, capacity, mechanism and peg context,
not as a ranked list of pools.

Scope boundary

Educational analysis only. APY is quoted
annualized APY, TVL is an observed proxy, and no
output is financial advice or a pool
recommendation.

Research questions

RQ	Question	Evidence
RQ1	How long do high-yield regimes last?	Median APY ≥ 10 episode duration: 2.0 days.
RQ2	Which mechanisms explain headline APY?	Base APY coverage: 90.3%; reward APY coverage: 61.6%.
RQ3	How stable are top-yield sets?	Churn rises from 13.5% to 33.3%.
RQ4	How does peg stress change APY interpretation?	Nominal APY is harder to read when the stablecoin denominator moves.
RQ5	How can pools be compared without ranking?	38 joint-screen candidates show trade-offs, not recommendations.

Reading order

The report moves from measurement to interpretation: duration first, mechanism second, stability and context after that.

Assessment logic

A strong answer is not a top-pool list. It is a transparent comparison that keeps uncertainty visible.

Design decision

The report answers comparison questions without creating a leaderboard. Each research question adds context that a single APY column would hide.

Data sources and feasibility

Source	Role	Evidence
DeFiLlama yields	Pool universe, APY components, TVL and history	15,669 live pools observed; 250 selected for full analysis
DeFiLlama stablecoins	Stablecoin metadata and peg-price context	404 assets and 2,012 price date records verified
CoinGecko Demo	Fallback checks for selected stablecoins	Keyless sample returned 3 assets on 2026-07-08
Protocol documentation	Mechanism labels and caveats	Classification is metadata, not a safety rating

Endpoint	Observed	Minimum evidence
Yields /pools	15,669 pools	APY, APY components, TVL, chain, project and pool ID verified.
Yields /chart	503 observations in sample	Historical APY and TVL are available at pool level.
Stablecoins	404 assets	Peg metadata and stablecoin price context available.

Verified source constraint

DeFiLlama is the primary source. CoinGecko is only fallback/enrichment because public historical depth can be plan-limited.

Raw preservation

Live responses are stored as request envelopes with checksums before transformation into canonical tables.

Pipeline and canonical schema

Step	Stage	Output
1	Collect	Pool, chart, stablecoin and fallback payloads.
2	Normalize	Timestamps, APY components, TVL and identifiers.
3	Resolve	Pool IDs, stablecoin exposure, protocols and pool types.
4	Validate	Critical, error and warning checks.
5	Analyze	Episodes, survival, churn, event studies and joint screening.
6	Render	Figures, PDF, presentation previews and manifest.

Artifact	Rows	Role
pools	250	Canonical entities
pool snapshots	137,095	Clean observations
stablecoins	394	Stablecoin metadata
stablecoin prices	176,176	Peg context
risk events	7,763	Event context
pool-day panel	137,095	Analysis grain
yield episodes	9,490	Episode analysis

Entity resolution

Pool IDs preserve the DeFillama source identifier; stablecoin exposure uses exact IDs and documented ticker overrides.

Canonical grain

The pool-day panel is rebuilt from processed tables, not from manual spreadsheet state.

Analytical grain

The central unit is a DeFi pool observed on one UTC day. This grain supports duration, ranking churn and event-time analysis.

Quality gates and metric definitions

12
checks

0
critical
failures

2
warnings

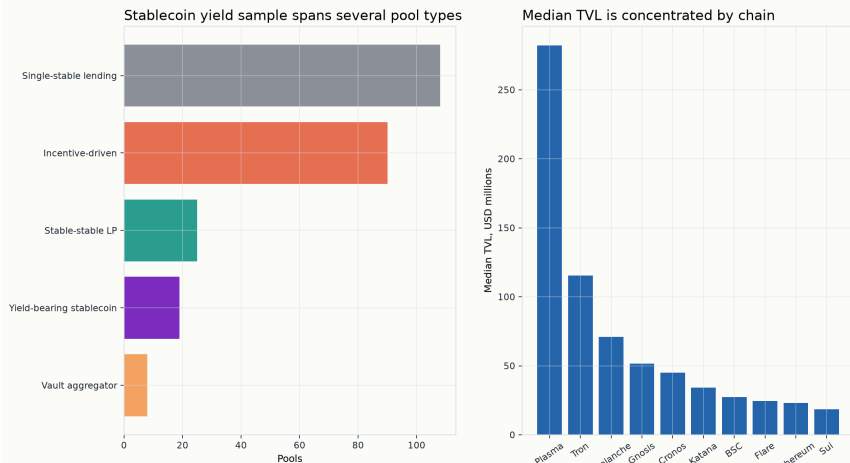
Check	Status	Failed	Metric	Definition
panel.exists	passed	0	High-yield episode	Continuous APY >= 10 percent run.
pool_day.unique_key	passed	0	Survival	Kaplan-Meier with censored active episodes retained.
pool_day.apy_total.completeness	passed	0	Ranking churn	1 minus top-k retention across horizons.
pool_day.tvl_usd.completeness	warning	26	TVL response	Median normalized TVL index around APY jumps.
pool_day.chain.completeness	passed	0	Frontier	Above median APY, persistence and TVL.
pool_day.protocol_id.completeness	passed	0		
pool_day.pool_type.completeness	passed	0		
apy.non_negative	passed	0		

Quality decision

Extreme APY rows are flagged, not hidden.
Removing them would make the
source-reported distribution look cleaner
than it is.

What is the observed stablecoin yield universe?

Yield universe: APY comparisons need category and capacity context



Sources: DeFiLlama, CoinGecko and selected official protocol documentation. APY is annualized quoted APY, not realized return.

What is the observed stablecoin yield universe? The sample covers multiple pool types and chains, so APY should not be compared without stratification. Sample: pools=250, pool-days=137095.

The sample spans multiple chains, protocols and pool types. A single APY number is therefore not a comparable market price.

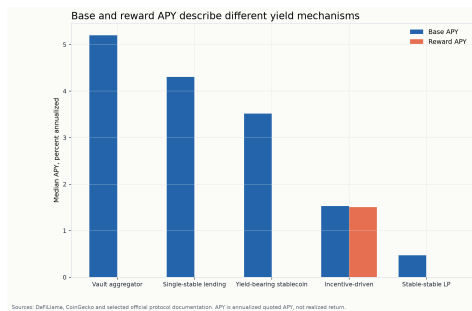
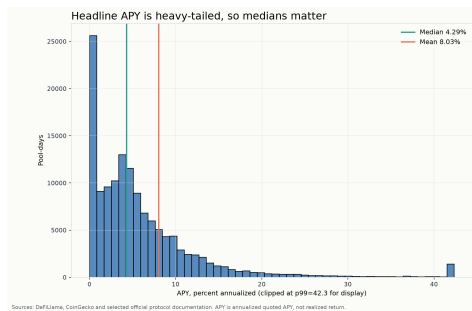
The final scope contains **250 pools** and **137,095 pool-days**.

Metric	Value
Pools	250
Chains	25
Protocols	84

Key reading

APY must be read with category and capacity context.

What creates headline APY?



Distribution and APY composition show why a single mean APY is a weak summary.

Pool-day median APY is **4.29%**, while pool-day mean APY is **8.03%**.

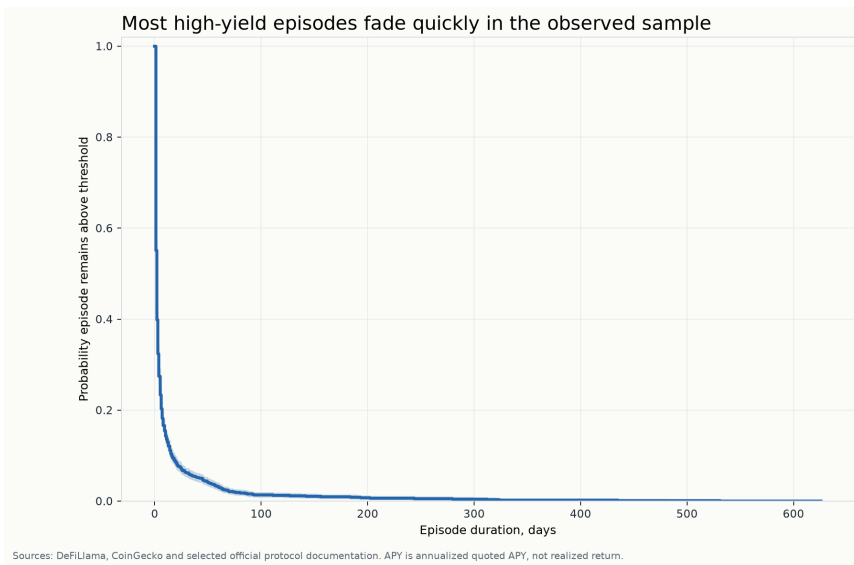
The gap indicates a heavy-tailed distribution: a small number of extreme observations pulls the mean above the median.

Base and reward APY describe different mechanisms. Reward-heavy pools should not be compared blindly with base-yield pools.

What it does not prove

It does not convert quoted APY into realized dollar return after fees, slippage or reward liquidation.

How long does high APY last?



How long do APY ≥ 10 percent episodes last? Kaplan-Meier survival declines steeply at short durations, with censored active episodes retained. Sample: episodes=2668.

The APY ≥ 10 percent definition produces **2,668 episodes**.

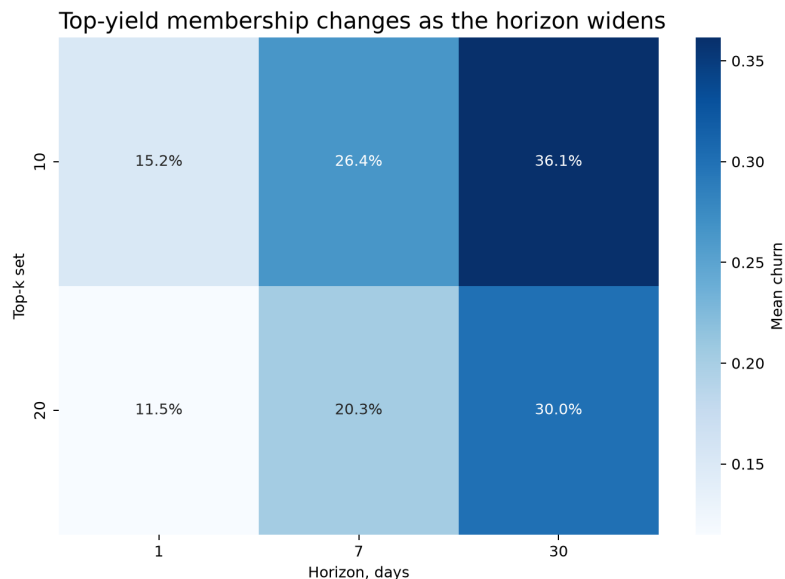
The median episode lasts **2.0 days**; by day 30, survival is only 6.2%.

Metric	Value
Median	2.0d
P75	5.0d
P90	16.0d

Key reading

High APY is usually an episode, not a durable pool trait.

Are top-yield lists stable?



Sources: DeFiLlama, CoinGecko and selected official protocol documentation. APY is annualized quoted APY, not realized return.

How stable are top-yield rankings? Heatmap cells show separate top-k means; pooled text summaries weight each valid observed-date/top-k comparison equally.
Sample: date comparisons=7938.

The top-yield set changes materially as the comparison horizon widens.

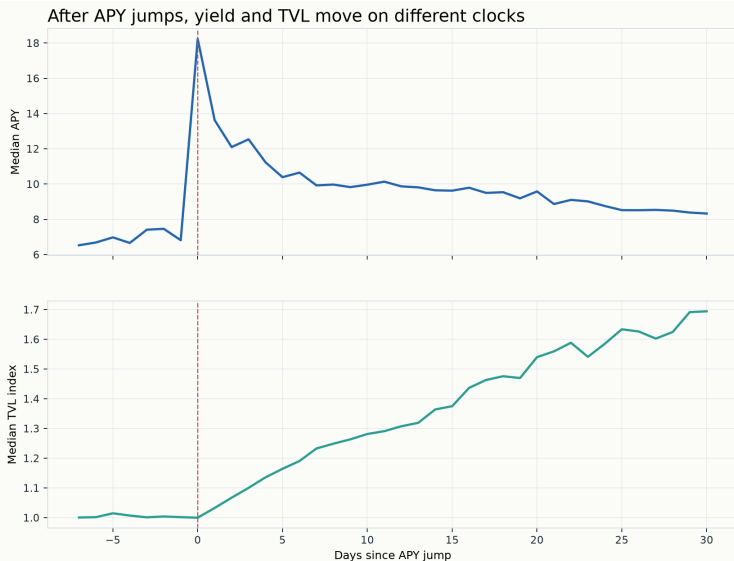
Comparison-weighted average churn rises from **13.5%** at one day to **33.3%** at thirty days.

Metric	Value
1 day	13.5%
7 days	23.6%
30 days	33.3%

Key reading

This is why the report avoids a best-pool leaderboard.

Do APY jumps and TVL move together?



Sources: DefiLlama, CoinGecko and selected official protocol documentation. APY is annualized quoted APY, not realized return. TVL index is an observed TVL proxy, not direct capital flow.

Are APY jumps followed by TVL changes and compression? Event-time patterns are observational and use TVL change only as a proxy. Sample: event-time rows=38.

APY jumps and normalized TVL response are aligned in event time.

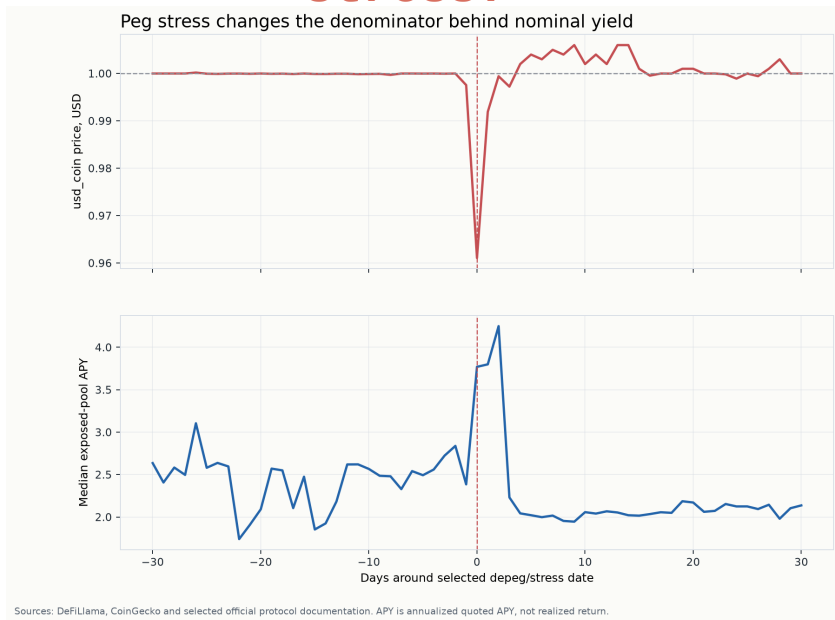
The pattern is observational: TVL can move because of deposits, withdrawals, price effects, accounting changes, migrations or source revisions.

Metric	Value
Events	453
Day -7	1.001
Day 30	1.694

Key reading

APY and TVL should be read on separate clocks.

What happens around peg stress?



How do APY and price behave around stablecoin peg stress? A depeg window makes nominal APY harder to interpret without price context. Sample: event-window rows=61.

The selected event is **usd_coin** on **2023-03-12**.

The minimum observed price is **0.9611 USD**, so nominal APY needs stablecoin price context.

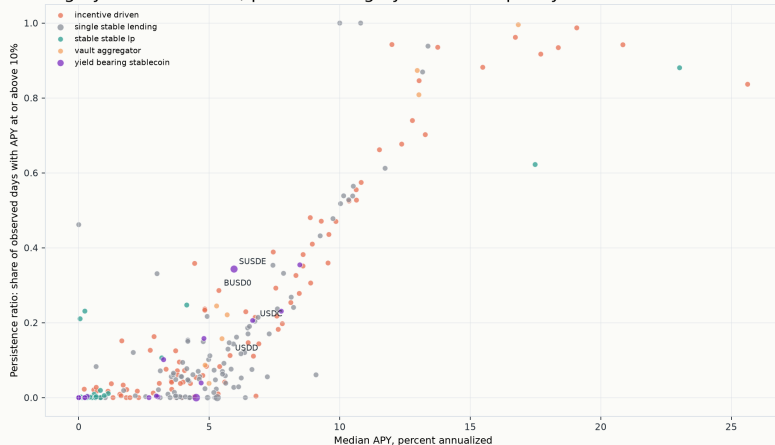
Metric	Value
Min price	0.9611
Peak dev.	0.0389
Pools	10

Key reading

Peg stress changes the denominator behind nominal yield.

How should yield be compared without ranking?

High yield is common; persistent high yield with capacity is rarer



Sources: DeFiLlama, CoinGecko and selected official protocol documentation. APY is annualized quoted APY, not realized return.

Which pools combine APY, persistence and TVL? Only a subset sits above median APY, persistence and TVL simultaneously; this is a joint threshold screen, not a recommendation. Sample: pools=250.

38 of 250 pools clear sample medians for APY, persistence and TVL simultaneously.

The joint screen keeps trade-offs visible instead of hiding them inside a score.

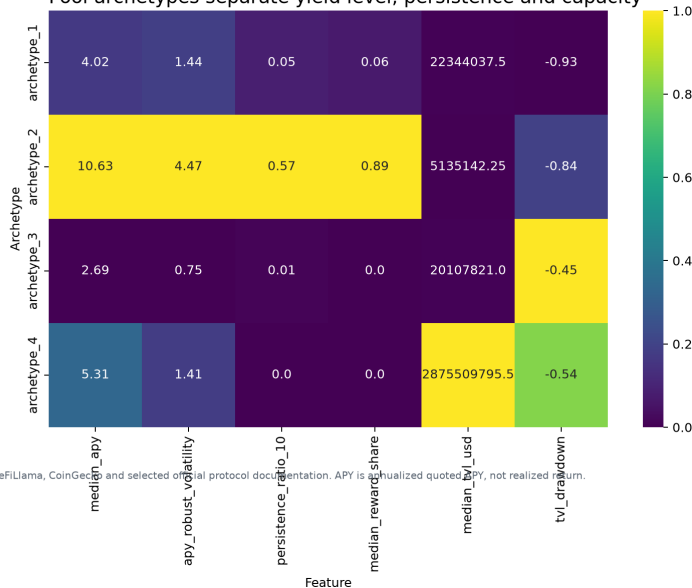
Metric	Value
Candidates	38
Share	15.2%
Pools	250

Key reading

The joint screen is descriptive, not financial advice.

Do pools cluster into interpretable profiles?

Pool archetypes separate yield level, persistence and capacity



Sources: DeFiLlama, CoinGecko and selected official protocol documentation. APR is annualized quoted APR, not realized return.

Do pools cluster into interpretable profiles? Exploratory clusters summarize patterns but are not labels of safety or quality. Sample: pools=250.

Archetypes summarize pool-level patterns across APY, persistence, TVL, reward share and volatility.

They are useful for discussion but are not risk labels, safety ratings or recommendations.

Metric	Value
archetype_1	138
archetype_3	61
archetype_2	48
archetype_4	3

Key reading

Clusters help describe heterogeneity; they do not replace the research questions.

Robustness checks

Check	Episodes	Median days	Censored
threshold_5	3,688	2.0	2.7%
threshold_10	2,668	2.0	1.2%
threshold_20	964	2.0	0.8%
min_tvl_500000	2,536	2.0	1.3%
min_tvl_1000000	2,470	2.0	1.3%
min_tvl_5000000	1,758	2.0	1.1%
min_history_90	2,668	2.0	1.2%
min_history_180	2,668	2.0	1.2%
min_history_365	1,919	2.0	0.7%
max_gap_0	2,683	2.0	1.2%
max_gap_1	2,668	2.0	1.2%
max_gap_3	2,646	2.0	1.2%
winsor_0.01_0.99	2,668	2.0	1.2%

Threshold sensitivity

Counts change, but median duration stays 2.0 days at 5, 10 and 20 percent APY.

Interpretation

Short duration is robust; it is not a universal forecast for every future episode.

Limitations and responsible interpretation

Concept	Boundary
APY	Quoted annualized APY is not realized return.
TVL	Observed TVL is not direct wallet-level capital flow.
Peg risk	Price deviations do not fully measure counterparty, legal, collateral or redemption risk.
Protocol risk	Mechanism labels are descriptive, not safety ratings.
Entity resolution	Bridged assets, renamed pools and migrations can remain ambiguous.
Visual ethics	The report avoids unsupported causality and recommendation framing.

No recommendation boundary

The project never identifies a best pool. Every pool-level label is descriptive evidence for the research question.

Reproducibility and deliverables

Task	Command
Install	<code>uv sync --extra dev</code>
Sample	<code>uv run python scripts/reproduce_all.py --mode sample</code>
Full	<code>uv run python scripts/reproduce_all.py --mode full</code>
Report	<code>uv run python scripts/build_report.py --mode full</code>

Deliverable	Path
Report PDF	<code>outputs/report/stablecoin_yield_report.pdf</code>
Report preview	<code>outputs/report/rendered_preview/</code>
Presentation	<code>outputs/presentation/stablecoin_yield_presentation.pptx</code>
Manifest	<code>outputs/release_manifest.json</code>

Final reading

High APY behaves like a regime, not a stable property. The joint screen keeps trade-offs visible without ranking pools.

Appendix A1. Tools and visual choices

The two visualization tools are **Matplotlib** and **Plotly**. They produce actual outputs. Seaborn supports Matplotlib and is not counted as a second independent renderer.

Tool	Concrete use
Matplotlib (+ Seaborn)	Ten static analytical figures in the report: distribution, survival, ranking churn, mechanisms, events and joint screen.
Plotly	Three interactive charts: survival with confidence band and risk sets, ranking churn, and threshold sensitivity.
pandas / NumPy	Canonical tables, grouped summaries and numerical transformations.
lifelines / scikit-learn	Kaplan-Meier survival / exploratory standardized pool archetypes.
Artifact Tool / ReportLab	PowerPoint assembly / report layout; separate from statistical estimation.

Why these forms? A step curve represents the discrete survival estimate. Its band makes pointwise uncertainty visible. The churn heatmap compares horizon and top-k size on a fixed 0-100% color scale with numerical labels. Threshold bars show how the episode count changes while the median remains two days.

Interaction with a purpose. Hover displays exact estimates, confidence limits and denominators. Range buttons reveal both short episodes and the full tail. Text summaries and HTML tables provide an alternative to mouse interaction and color perception.

Offline companion. Open **stablecoin_yield_interactive.html** from the revision directory. Plotly.js is embedded. The file requires no account, API key, CDN or internet connection.

Installed versions: pandas 2.3.3, numpy 2.4.6, matplotlib 3.11.0, plotly 6.8.0, lifelines 0.30.3. The complete Python environment is pinned in uv.lock.

Appendix A2. Reproduction and rights

<https://github.com/Laimon99/stablecoin-yield-visualization>

Published aggregate evidence, without APIs

```
uv sync --frozen --extra dev
uv run python scripts/reproduce_published.py
uv run ruff check src scripts tests
uv run pytest
```

The script reads the public survival, ranking and robustness tables and the report summary. It checks counts, survival ordering, confidence limits and headline consistency before regenerating the interactive output. The audit JSON records input hashes and installed versions.

Recollection and full method

```
uv run python scripts/reproduce_all.py --mode full
```

A new clone collects provider data. An existing checkout can reuse cached responses; `--refresh` requests recollection. Live revisions and the current eligibility sample can change results. Run in a separate checkout to retain the historical outputs.

Reproduction boundary. Rebuilding aggregate figures is distinct from independently reproducing raw-to-result transformation. The historical raw and row-level provider snapshot is not publicly redistributed. The optional PowerPoint builder also requires Artifact Tool. The rendered PDF and Python evidence companion do not.

Licensing. Code, tests and configuration: MIT. Original report, slide text, documentation and authored visual expression: CC BY 4.0. Credit Simone Ragusini, Stablecoin Yield (2026), link the licence and repository, and identify changes. These grants exclude provider data and third-party software.

Sources. DeFiLlama: [API documentation](#) and [terms](#). CoinGecko: [API documentation](#) and [terms](#). Plotly.js retains its embedded MIT notice. Licence scope: LICENSE and NOTICE.md.

Appendix A3. Statistical interpretation

Survival at day 30. $S(30) = 6.22\%$, with published 95% pointwise interval 5.34% to 7.19%. $S(t)$ means the probability that an observed episode lasts more than t calendar days. The two-day median remains unchanged.

Dependence. The lifelines interval treats episodes as independent. Multiple episodes from one pool can be dependent. This interval is neither pool-clustered nor a simultaneous band. Pool-level resampling would address a different inferential question and is not claimed as part of these results.

Observation boundaries. An episode already active when a pool first appears can have an unobserved earlier start. A long gap terminates an observed run under the original definition; it is not evidence of an economic threshold crossing. The gap sensitivity results support the stability of the observed median but do not establish non-informative censoring.

Selection. The sample deliberately combines high-TVL and high-current-APY pools after eligibility filters. It is not a random market sample. The 90-day and 180-day history checks apply within pools already selected with at least 180 provider observations, and cannot recover excluded short-history pools.

Ranking denominators. Headline churn averages valid date-by-top-k comparisons: 2,669 at one day, 2,657 at seven days and 2,612 at 30 days. These are not independent pools. Entry, exit and changing availability can affect membership.

Event response. APY jumps select unusually high values, so subsequent declines can also reflect regression to the mean. TVL can vary through prices, accounting and capital movements. Available events can differ by offset. Without a control group or causal identification, the curves support descriptive timing comparisons only.

What remains supported. The selected historical sample has short typical high-yield episodes, increasing membership churn with horizon, and substantial contextual differences between nominal APYs. These results motivate transparent multidimensional communication rather than investment recommendations.